LESSON 5.4

Apply Area Concepts to Solve Problems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a composite figure by decomposing it into basic shapes, including a regular polygon split into triangles, and adding or subtracting the areas.

WHY THIS MATTERS Your architecture firm is calculating the floor area of an L-shaped room for a client's new home. The contractor needs the total area to order the correct amount of flooring. The room can be split into two rectangles: one is 12 feet by 8 feet and the other is 6 feet by 5 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Composite Figure *Spanish: Figura compuesta*	A shape made by putting two or more simple shapes together.	triangles joined together
Decompose *Spanish: Descomponer*	To break a shape into smaller, simpler shapes.	hexagon → 6 triangles
Add *Spanish: Sumar*	Add up the areas of the smaller shapes to get the total.	T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft → total = 30 + 28 = 58 sq ft
Subtract *Spanish: Restar*	Take away the area of a missing piece from a bigger shape.	A 14×10 pool with a 6×4 cutout: 140 − 24 = 116 sq ft of water surface
Formula *Spanish: Fórmula*	A math rule written with symbols.	Rectangle: A = l × w; Triangle: A = ½ × b × h; use the right formula for each piece of a composite figure
Regular polygon *Spanish: Polígono regular*	A shape whose sides are all the same length and whose angles are all the same size.	stop sign, hexagon tile

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Regular pentagon: sides 10 cm, apothem about 7 cm.

1. 5 sides make 5 triangles.
2. One triangle: ½ × 10 × 7 = 35 cm².
3. Five triangles: 5 × 35 = 175.

Answer: Area ≈ 175 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now an L-shape made of a 5 ft by 4 ft rectangle and a 3 ft by 2 ft rectangle.

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I decomposed the figure into _____ congruent _____ .
- The area of one triangle is _____ square units.
- There are _____ triangles, so the total area is _____ .
- I know the triangles are congruent because _____ .

Word bank regular polygon | decompose | congruent | triangle | center | base | height | multiply | total | area | square units | composite figure

WATCH OUT!

A common mistake

A common mistake in Area of Composite Figures is adding every piece even when a piece has been cut out of the figure. For a 10 ft by 8 ft rectangle with a 3 ft by 2 ft rectangular notch cut from one corner, students compute the full rectangle 10 × 8 = 80 sq ft and stop, or they add the notch instead of removing it (80 + 6 = 86 sq ft) instead of subtracting it: 80 − (3 × 2) = 80 − 6 = 74 sq ft. Before submitting, check whether each piece is part of the figure (add it) or missing from the figure (subtract it).

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Answer: Area ≈ 175 cm²

1 **Circle the letter of the best answer. Show how you know.**

An L-shaped room is made of two rectangles: 10 ft × 6 ft and 4 ft × 3 ft. What is the total area?

(A) 12 sq ft
(B) 60 sq ft
(C) 72 sq ft
(D) 72 ft

STEPS TO TRY

1. The L-shape is already split into two rectangles for you: 10 × 6 and 4 × 3.
2. Find each rectangle's area separately with A = l × w, then combine — joined shapes mean adding.
3. Compute 10 × 6 and 4 × 3, then add the two results. Label your answer in sq ft.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A rectangular patio is 15 ft × 10 ft with a 5 ft × 4 ft rectangular flower bed cut out. What is the remaining area?

(A) 20 sq ft
(B) 130 sq ft
(C) 150 sq ft
(D) 170 sq ft

STEPS TO TRY

1. The flower bed is CUT OUT of the patio — that phrase tells you which operation to use.
2. Find the full patio area (15 × 10) and the cutout area (5 × 4), then remove the cutout.
3. Compute 15 × 10 and 5 × 4, then subtract the flower bed's area from the patio's area.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

(A) 8 sq units
(B) 15 sq units
(C) 20 sq units
(D) 23 sq units

STEPS TO TRY

1. Two rectangles make this L: one is 5 × 3 and the other is 2 × 4.
2. Joined pieces mean you add their areas after computing each one with length × width.
3. Find 5 × 3, then 2 × 4, and add the two products together.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

When should you subtract areas instead of adding them to find the area of a composite figure?

(A) When the shapes overlap
(B) When one shape is larger than the other
(C) Always subtract — never add
(D) When a piece is cut out or removed from a larger shape

STEPS TO TRY

1. Think of real examples: a window in a wall versus two rooms joined into an L-shape.
2. Ask for each choice: is material missing from the shape, or are pieces being combined?
3. Picture a wall with a window: you can't paint the glass, so its area comes OFF the wall's area. Match that idea to a choice.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

SENTENCE STARTER I ___ the areas because the composite figure is made by ___ a piece from a larger shape, so the total area is ___ - ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help